

ChatGPT in the English Language Arts Classroom

Introduction

Doing things with words is a large part of our educational process: it's through writing in which we develop, refine, and reflect on our ideas and come to constitute knowledge. Writing, of course, is a very ancient technique - a technology beleaguered by the forces of time - always undertaking alteration: perpetual metamorphosis. What necessarily follows is that what it means to write, what it means to read, and as an extension of those two exercises, what it means to think and to learn, are permutable, always enmeshed in an updated technique or technology. I consider this paper's contributions to be a record of an extended inquiry into this claim by route of the English Language Arts (ELA) Classroom. ~~My contention is that as writing and reading become increasingly enmeshed with AI, and for the purposes of this paper, specifically ChatGPT, so will our ELA curriculums and classrooms.~~

The project seeks to trace the origins of ChatGPT, and therefore to a minimal extent, Artificial Intelligence, with the goal of mapping its historical migrations onto the developing territories of automated or augmented deep learning systems. These systems encompass and enforce the trajectory of the tool and its use in an educational setting. The corollary discussion that follows addresses primary concerns into how the tool is being used and misused throughout secondary education English Language Arts classrooms in the United States, and subsequently examining the barriers or challenges to entry faced by both students and teachers. The rest of the paper dedicates a section to perspectives of learning and motivational theories as they relate to ChatGPT; thereafter considering the tools implications for Universal Design for Learning (UDL), which aims to provide an exhaustive index of the "digital divides" that complicate the intended use of these tools while also addressing critical issues of accessibility. Our primary research questions are influenced by thinking through AI as a tool for

experimenting with literary features and supporting the writing process within an ELA setting. We want to ask the right questions: How does this tool work? What is its history? But most importantly, what are the implications of ChatGPT, particularly on the ELA classroom in secondary education, and what then, as an educational tool, does the platform make feasible for the future of literacy in the realm of English Language Arts? What does it foreclose on English educators? Language Arts learners? What, ultimately, is the transformative potential of ChatGPT on ELA education?

What Is ChatGPT?

My project takes shelter in the aftermath of the anchoring of thought in an archive –the capturing, preserving, and disseminating of words into material systems. From the advent of writing to the revolution of the printing press, onto the typewriter and its uncompromising successor, the word processor, onwards to the computer and the internet, each step has reshaped how we technologize the word. Now, trailing at the rear we encounter our interactive partner-in-crime: Generative AI, and more importantly for our purposes, ChatGPT. Our question becomes what does ChatGPT *do*? How can we make its functionality visible so as to see how the tool's capabilities and limitations may provoke or intervene in English Language Arts classrooms? We can understand the tool through the following introduction, provided by its creator, OpenAI (2022):

“ChatGPT (Generative Pre-trained Transformer) is a conversational AI that interacts through a dialogue format, enabling the prompter to ask questions and seek information. It draws on extensive pre-training and fine-tuning of data to generate coherent and contextually appropriate responses” (Introducing ChatGPT, 2022).

We have entered the world of word and image prediction AI. This “fine-tuning” of data mentioned above is enabled through *data-driven decisions*, or by a process called deep learning. ChatGPT demonstrates the capacity to fine-tune, or execute tasks by way of “identifying and extracting ‘patterns’ from large datasets that accurately map from sets of complex inputs to good decision outcomes” (Kelleher, 2019). In that way, this large-scale autoregressive language model, with over 175 billion parameters, is capable of in-context learning; to provide a response without the necessity of task-specific fine-tuning (whereby an LLM is trained on a variety of concrete examples) (Brown et al., 2020). It does so through the concept of a function, or by deterministic mapping from inputs to outputs (Kelleher, 2019). Functions are central to the tools learning process, and have a complex representation through neural networks – ‘patterns’ that deep learning algorithms extract from datasets; that is, neurons in an overarching network, made up of simpler functions that act in relation to each other, and the overall (more complex) function, defined by the network (Kelleher, 2019). Kelleher’s characterization of these nets as “pattern-detectors” could be challenged as an assumption – patterns are not received holistically or conceptually, but rather data inputs are treated as indexes, fed forward and then back propagated through the net in an indexical process of forming connections between informational units (Weatherby & Justie, 2022). However, for our purposes we will use the term.

Deep learning rests on the power of machine learning to pursue the best function to match the patterns in the dataset to a particular output. An extended training or pre-training process allows a neural network to point at features and identify the function (model) that best matches a particular pathway. In an iterative process – adjusting its model parameters to minimize area – the net arrives at the classification stage codified in the net’s output layer. Thus, rather than any intrinsic understanding of a certain classification, nets are able to classify canines, for example, without having an intrinsic understanding of canine

classification. Once trained, the model is applied to new, unseen data to generate more predictions or classifications, or to “generate [more] coherent and contextually appropriate responses” (Weatherby & Justie, 2022).

What Existing Technologies Does This Adopt?

What this description and overview (*What is it? What does it do?*) underscores is ChatGPT's ability to manipulate large volumes of data. In addition to the billions of parameters in which these AI models are built, there are also billions of words generated by AI in the consumer marketplace, as the technology has become a core part of global capitalist infrastructure. It is used for targeted advertising, pricing systems, social media, content delivery, facial recognition, and more (Weatherby & Justie, 2022). The time is now ripe for language-based, conversational AI, as deep learning has ushered in breakthroughs in natural language processing (NLP). With advanced NLP, AI generative systems are able to straightforwardly assist with text generation, error correction, and content refinement (Eisenstein, 2019). In a 1950 essay titled “Computing Machinery and Intelligence,” a machine was deemed to have passed the Turing Test if it could engage with human interlocutors in natural language. Thus, the idea of an intelligent machine has always been linked vis-à-vis human capabilities for language. It is no surprise then that the advancement of natural language processing emerged as a primary aim within the field of computer science (Hayles, 2012). We can track more direct manifestations of ChatGPT back to the 1960's, with features such as the ELIZA program - an AI cast as a psychotherapist through the mind of MIT professor Joseph Weizenbaum (Erickson, 2023). The software was relatively simple: user inputs were applied to a set of rules and plausible responses were generated by the program as a result. User would type their input on an electric typewriter connected to a mainframe. After a moment, the program - or ELIZA - would reply (Tarnoff, 2023). Many of the ethical and social questions posed by this technology at the time, including those Weizenbaum made himself, still stick with us today and influence

the research and conversation around this tool in the ELA classroom. Other features or precursors include the “transformational technologies” of the 2010s; that is, virtual assistants like Siri, Cortana and Alexa, all of which received widespread commercial adoptions through the marketplace; the term “virtual assistant” also raising interesting questions about the purport of our tool along educational lines. More so, the tool is also enacted through technologies like spellcheck, built-in dictionaries or thesauri, Grammarly, or what we should more abstractly call predictive text interfaces - are all examples of writerly tech assistance; all types of technologies that allow for some form of, as Alan F. Blackwell puts it in his book *Moral Codes*, “Chatting to the Stochastic Parrots” - lifeless models which process text as statistical sequences (Blackwell, 2024).

Infrastructure

Just as the patterns of error in writing are influenced by the pen or the typewriter, so too are those shaped by a word processor, where the keyboard acts as an extension of thought, rather than a mere external device (Hayles, 2012). With an anthropomorphic understanding of the tool and a partial view of its historical lineage (so as to avoid complete technological essentialism), it becomes crucial to examine ChatGPT's role in this latest phase of technologization, especially as it embeds itself into educational infrastructure at a superwarped speed. ChatGPT necessitates some of the same functionality as other artificial intelligence tools (think QuillBot and Grammarly). It is an accessory to the writing process, a means to paraphrasing, to summarizing, offering grammatically correct stylistic enhancements to improve textual clarity (Raheem, Anjum, & Ghafar, 2023). The scalability of these products stems from subscription-based or freemium business models that rely on widespread adoption and a demand for personalized writing assistance, “paving the way for a more proficient and streamlined approach to English academic writing” overall (Raheem, Anjum, & Ghafar, 2023). ChatGPT and adjacent writing assistance tools are predicated on their

ability to enhance efficiency. What a business model does – what scalability does – is structure the space to which a tool belongs, so much so that that structure becomes the register in which a tool can be interpreted and put to use. Consider, for example, the materializing evidence surrounding AI technologies (in this case Grammarly and ChatGPT) accelerating language change by promoting their particularly inert and homogenizing prose (Shien, 2024) (Rudnicka 2023). Is this an early lesson to the problem of scale, or learning at scale, with this tool?

In an effort to sever ChatGPT from its writing assistance lineage, it must be said that we are dealing with a technology that may underpin 7 to 10 percent of global GDP within a decade, implying the potential for vast growth across industries, with Microsoft and OpenAI accounting for 69 percent of the total market share for generative AI models (Vipra & Korinek, 2023). The same cannot be said for tools like Grammarly or Quillbot, who do not own proprietary GPT technology at the level of OpenAI. The implications of our tool are vast, and their development expensive: such models requiring substantial fixed costs – large expenses on computational power as well as the skilled talent to train large datasets (not to forget about the outsourced workers who label data and help train models via reinforcement learning from human feedback for egregiously low pay) (Vipra & Korinek, 2023). In addition, the market for the computer chips that are used to train and deploy foundation models is highly concentrated, and the costs of manufacturing additional equipment for producing computer chips is high, as they rely on highly specialized manufacturing techniques and high fixed costs (Vipra & Korinek, 2023). Vipra and Korinek's study also reveals why this may change:

“The high returns generated by the scarcity of computer chips as well as government subsidies provide strong incentives for investment in additional capacity and may help to diversify the semiconductor supply chain and bring down costs.”

Nevertheless, a key distinction to attend to is the difference between the sequence-prediction capabilities inherent to these large language models (LLMs) and the manner in which these capabilities are utilized to construct LLM-based chatbots. While the transformer-based LLMs focus on predicting and generating coherent sequences of text based on patterns in their training data, LLM-based chatbots supplement their “coherent and contextually appropriate responses” with a dialogue management system (DMS). This additional layer of infrastructure allows for the incorporation of prompts, context management, and predefined conversational frameworks, which adapt the raw sequence-prediction functionality for interactive use; it is this system that has given ChatGPT its huge commercial value (Trevisan et al., 2024).

Education as an industry may lag behind in wide scale adoption of this tool, but most studies on ChatGPT and its purport for education aim for the time after contemporary hype, where we could consider the significant opportunities and challenges posed by its application within educational infrastructure. The sheer scale, the amount of computing power, and the DMS “role playing” allow this tool to fulfill diverse roles—including serving as tutors, tutees, learning partners, domain experts, and administrators—thereby providing unsurmounted contributions to educational systems. That, at least, is what we are told through a rather aspirational Artificial Intelligence in Education (AIED) framework advanced by more recent applicational research (Gwo-Jen Hwang & Nian-Shing Chen, 2023).

The formal conditions of ChatGPT described above are shaped and hardened by issues concerning what some researchers call “pathologies of scale” – or, the lack of regulation, driven by industry narrative and corporate control, around the development and deployment of this tool (Kak & West, 2023). All of which, of course, will entangle and refract within the educational system in myriad ways. Most notably, the significant barriers to participation faced by small businesses or startups – given that only a handful of companies with incredibly vast resources are able to build GPTs – means that a few big tech companies will control,

dominate and enshrine themselves into the space, perpetuating a type of dependency on these firms (Kak & West, 2023). The question then becomes: how will this tapestry of assemblages trigger new transformations and proliferation within education moving forward? More specifically for our use, what does the lack of diverse innovation mean for how this tool is set to emerge within the space of the ELA classroom? How may it influence pedagogical decisions about English Language Arts? Even more troubling, perhaps, are the many existential harms associated with these LLM's (Kak & West, 2023). AI's largely invisible infrastructures (which I have tried to make note of) point to four core areas of concern; concepts that will be introduced here and developed in relation to the ELA classroom moving forward. Before ChatGPT can dissolve any of the stubborn challenges of education, its logical and practical limitations must be examined. These include its biases and exclusions, its environmental impact, its approximations, illusions of understanding, and lastly this centralized concentration of power over resources, (Bender et al., 2021) all of which are barriers or challenges we should keep in mind as we venture into use and (misuse) of this tool in English Language Arts classrooms.

ChatGPT in Secondary ELA Classrooms

Now that we have attempted to demystify ChatGPT as a technology, we can now feel comfortable demystifying it as a learning tool. What new possibilities might be emerging from this evolving space? It'll be important to keep in mind the aforementioned expanse we have before us. This research begins with the definitional implications of the cognitive process model on writing, which breaks down writing into three constitutive parts components: the task environment - exterior forces influencing the writer (rhetorical problem, the topic, audience), the writer's long term memory, and the recursive process of planning, translating, and reviewing. The problem-solving nature of this goal-directed theory of writing was that it was exactly that: goal-directed. By way of its own articulation, it created limitations in addressing the social and cultural dimensions of writing (Zimmerman, 1998). Just as

important as the model itself was the critique. In response, Flower adapted her framework, resulting in a social cognitive theory that embedded writing within historical, social, and rhetorical contexts. Similarly, Beck and Levine (2024) situate the use of ChatGPT in writerly communities, including the social, cultural, and institutional dimensions that influence the writing process. We will now inspect both theories – the social cognitive theory of writing and its predecessor – as they relate to our tool and its effect on the ELA secondary education setting.

Much of Levine's research investigates how students leverage ChatGPT to plan, translate, and review their writing; the degree of scholarly attention focused on writing primed me to be especially curious to this line of inquiry: how do secondary students react to the tool in the face of writing or in the exploration of a topic (Levine et al., 2024)? Levine's research maintains how the tool has been used thus far as well as its ideal use. Ideal use for secondary education students is prompting the chatbox during the initial planning stage where students generate ideas, set goals, and organize (Levine et al., 2024). Melissa Libertino's work complements this claim especially well. As both a researcher and an 11th and 12th-grade teacher, she adopted a more personalized approach for some of her ELA students who needed help outlining their writing. More particularly, the tool was implemented to encourage tailored outlines specific to assignments and, more importantly, to individual needs. Students input prompts to specify grade level, writing style, thesis requirements, and desired paragraph structure (Libertino, 2024). Moving forward, Levine's research suggests that the tool is less likely to be deployed in translating – or, in “moving from ideas in the head to generating actual text on a page or screen,” but that, during this stage, ChatGPT has the capabilities to advance multiple, tailored, contrasting translations of the same idea, which allows for the emergence of what Levine calls “meta-level understanding;” were students able to identify the common features of X, whatever X might be, without the need of a teacher (Levine et al., 2024). About one-third of prompts involved reviewing: editing and evaluating

drafts, with some students incorporating ChatGPT's suggestions verbatim, while others selectively integrating its edits. Students often determined their argumentative stances before consulting the tool, with only a few confirmed cases of shifts in position after interacting with ChatGPT (Levine et al., 2024). In addition, the students managed the bot's contributions through precise prompting, "setting the goals that shape the collaborator's influence, rather than the other way around" (Beck & Levine, 2024). The Preliminary work on ChatGPT in the ELA classroom undertaken here states that students often ask ChatGPT to take on a knowledge-persona or to provide a good starting point for various tasks, such as drafting or brainstorming. It excels at offering structure, helping users organize their ideas effectively. Students see models or examples and transfer new knowledge to their own writing. For Levine, this is enough to agglomerate ChatGPT into the social process of writing distributed across teachers, peers, settings, histories, and technology. More than that, though, is the question of exertion: when the student writer lets go of control and enables the tool to be more involved in the writing process, do the biodiversified social and cultural dimensions of writing reduce down to the rhythm of statistical frequencies? To combat this, Levine has the antidote: "what ChatGPT can't do is become less like a fellow writing collaborator, and more like an enforcer of historical and cultural influences on written discourse" (Levine et al., 2024). Importantly, the research is backed on the basis that unquestioning acceptance of ChatGPT's output was not the norm among high school participants (Beck & Levine, 2024).

The school in which this research was conducted provided an unusual level of academic support, which may not be representative of a typical high school setting or the usual way this tool is being utilized across a vast spectrum. For one, prompts often require clear specifications and clarifications to yield the desired results. While ChatGPT can handle creative tasks, its responses in these areas tend to be more speculative, lacking the grounding it can achieve when addressing the more recursive processes of planning, translating, and reviewing. Barriers or challenges then have to do with the perceived potential of the tool and

how the tool might be misused. Are educators like Levine overinterpreting the strings of symbols produced by GPTs, anthropomorphizing AI, and inadvertently succumbing to the "Eliza effect" -the tendency to project human traits onto computer programs? One way of understanding this is through our analysis of the DMS of ChatGPT, which presents the statistical sequence-generating capabilities underlying the LLM as a conversation rather than a simple series of probabilistically-chosen words (Trevisan et al., 2024). The tool is being misused in the sense that there is no intrinsic cooperative dialogue, and therefore no pedagogy, no advancement of knowledge at its core. Misuse oftentimes comes from the models' misplaced approximations (Goodlad & Stone, 2024).

Another logical limit is the tool's inability to predict and/or recognize a new anomaly, which renders the chatbot incapable of producing new metaphors in a consistent or non-random way. There is never the invention of codes and worlds, only the reproduction of vector spaces that make up the model's parameters (Pasquinelli, 2019). What's more, research into "contextual noncompliance in language models" is interested in the instances where a student may make incomplete or unsupported requests, indeterminate or subjective inquiries, and altogether prompts containing false information. GPT-4, rather unfortunately, frequently fails to appropriately identify these nuanced scenarios, with the model complying incorrectly up to 30% of the time on unsupported prompts (Brahman et al., 2024). Thus, education will have to reckon with these "bullshit" language games that prioritize plausibility over factual accuracy (Trevisan et al., 2024), and seriously think about what this means for the social, cultural, and institutional dimensions that influence the ELA classroom and the writing process.

Some of the research I encountered reflects the prevalent hype - or perhaps fearmongering - propagated by OpenAI's business model, elite circles, and media narratives, which often frames LLM-based chatbots as entities possessing intention, knowledge, and reasoning capabilities. As

OpenAI chorales the chorus of its intended user-base in our ever-expanding efficiency boom, teachers have to confront significant information gaps, as many lack the overall knowledge or expertise necessary to introduce ChatGPT's capabilities to specific learning tasks or within the needs of diverse learners. Yet, teachers have needs of their own. At the moment, vague and irrelevant guidance and a lack of resources addressing classroom specific applications and pedagogical objectives keeps teachers constrained by misconceptions, or perhaps genuine hesitations of distrust (Tan & Subramonyam, 2024). Information gaps further faulty mental models, as teachers increasingly turn to social media as a primary source of information (Tan & Subramonyam, 2024). Teachers often revert to generating basic outputs because of their constrained abilities around prompt engineering, but also because of constrained time, which impedes barriers to entry and integration. Consequently, the evolving dynamics around AI literacy in the ELA classroom -- the examination and the opinions on what constitutes learning versus cheating in writing tasks, or the integration of ChatGPT into classroom practices more generally - falls around perceptions of the tool by both teachers and students.

~~In some of our aforementioned research, ChatGPT provided unique advantages that sincerely helped students recognize and refine their writing styles. Nonetheless, its role in revising -- often students taking it as an all-or-nothing approach to their writing -- points to areas where more nuanced instructional strategies are needed (Beck & Levine, 2024).~~

With the advent of ChatGPT into an educational context, students and teachers have conferring and competing perceptions over the perceived potential of the tool, its actual use in a learning context, and its potential for misuse. Seeking an explanation, a more recent study (Mah et al., 2024) asked a group of students and a group of teachers for their perspective on learning and cheating with ChatGPT. In the study, both groups arrived at similar conclusions visible through the production of similar criteria in determining their rankings of hypothetical ChatGPT use cases. Themes included how often ChatGPT provided accurate models for

knowledge-making, as well as accurate examples, alternative perspectives, and furthermore its ability to attend to corrections. In both the student and teacher groups, attention centered around the idea as to whether the student or the Chatbot was doing more of the thinking, with teachers referencing context – contextual factors such as assignment goals, or what the student did with a given output – at twice the rate (19.5%) as students (8.6%) (Mah et al., 2024). Other tensions that emerged between the two perspectives had to do with the enabling of shortcuts – students bypass knowledge acquisition – versus actual scaffolding for deepened learning. These tensions also included employing ChatGPT for idea generation (prompts, open sentences, counter-arguments) versus language generation (full sentences), and how often ChatGPT's assistance reflected traditional resources of support like teachers or textbooks (Mah et al., 2024).

Technological leaps require new forms of literacies, yet schools often lack the resources, personnel, or capabilities to keep pace with change. Teachers and students rarely participate in developing policies and norms for adopting new learning technologies although they are the most directly impacted by them. Against this backdrop, there is a growing necessity for teachers and students to construct shared norms around responsible use of AI (Mah et al., 2024). The research above highlights the importance of engaging both teachers and students in co-constructing norms and policies around responsible AI use in educational contexts, as well as the need for administrative support systems, tools to facilitate responsible integration, and an overarching educational framework emphasizing AI literacy. If this tool is to be seriously implemented, pedagogical practices should service a student's recognition of the historical and cultural biases embedded in ChatGPT's outputs. If anything, these limitations reinforce the importance of students maintaining their agency, operation, and learning purposes when using the tool to complete writing tasks. Students should lead their interactions with AI, and must still develop a discerning ear for “bad metaphors” by expanding their understanding of style, theme, genre by reading human-authored texts (Beck & Levine,

2024). Going forward, Teachers face logistical and ethical challenges, such as addressing academic integrity, managing classroom procedures, and adapting assessments to accommodate AI use (Tan & Subramonyam, 2024). However, teacher intervention should guide students in using ChatGPT responsibly, emphasizing meta-awareness and critical engagement with AI outputs.

Learning Theories

The challenges of ChatGPT complicate and extend onto the traditional challenges of reading and writing well. Writing-to-learn implicitly involves students grappling with the rules, conventions, strategies, and textual genres of writing, meaning that both writing and learning are interconnected, as Linda Flower asserts through her constructivist view of the writing process (Zimmerman, 1998). The implementation of learning and pedagogical theories provides a framework for understanding how ChatGPT integrates into educational settings, specifically in the context of the ELA classroom. Constructivist learning theory, grounded in the works of Piaget, emphasizes the active role of learners in constructing their own understanding through experience and problem solving. Piaget argued that knowledge acquisition was the result of not only perception but also practice. The combination of two competing theories of knowledge at the time – the empirical school and the preformation approach -- laid the foundation for a novel combination and eventually the wide scale application of the constructivist learning theory (Hof, 2021). Changes in the paradigms and expectation of science mirrored changes in culture and mentality – constructivism thus can only be understood against the historical backdrop of the mid-twentieth century, and more specifically the fervent countercultural movements of the 60's (Hof, 2021). In turn, historical circumstances laid the groundwork for the research that was to be done at MIT, where Marvin Minsky and Seymour Papert first began to conceptualize artificial intelligence as a process imitating human cognitive development. Hovering over Piaget's theories of childhood development, they sought to design a computer system that could progress from simple to

complex knowledge structures, paralleling the child's learning process. "In other words, if it was accepted that a child learns by construction - understood as a process of building knowledge structures - why should the computer not be able to do the same?" (Hof, 2021). Paper's extension of Piaget's constructivism theory of learning is subsequently extended onto a technological artifact, or done *through* the creation of an artifact - in this sense learning comes to be seen as something we do *with* technology (Selwyn, 2017). These scientific epiphanies, along with shifts in policies and nation-wide research reforms, instigated developments in computer-based education technology: Papert's Logo programming language. Papert's division of research, backed by a research group, with the likes of Minsky and computer scientist Cynthia Solomon was dedicated to not only "new roles for the computer in education," but also curating better theories concerning human cognition (Hof, 2021). It is here where constructionism is born - combining the learner with his/her medium (Hof, 2021).

I mention the history briefly here in recognition of the shared overlapping between both the where and how of the constructivist and constructionism models of learning and the, what we can consider, bare carcass of ChatGPT: the beginnings of artificial intelligence research. In trying to better understand human thought and learning processes, people like Minsky, Solomon, and Papert were also trying to build better machines. The history is important, as it lets us think through how Levine or Liberto's research subjects extend their writerly abilities through active creation and integration with an AI technology. By making use of an object in their environment, the students enable themselves more sophisticated thoughts than would otherwise be possible. Or, as Levine puts it in her conclusion, students use the technology "to seek feedback that informs and improves their practice" (Levine et al., 2024). Much like the process of "writing down" as an integral aspect of thinking, research into the interplay of constructionism and artificial intelligence in the classroom states how students are encouraged

to explicitly think about *how* they solve problems (Kahn & Winters). Students “construct” a public entity, like a paper or an essay, by utilizing previously acquired concepts from ChatGPT or from previous pedagogical processes. Rhyming with the social cognition theory of writing, students ease themselves into a task environment by considering, for example, how to write an opening statement; later moving onwards to the recursive processes of planning, translating, and reviewing, eventually arriving at an artifact, all through the use of an extended sociotechnical medium (Beck & Levine, 2024) (Zimmerman 1998).

Hwang's Artificial Intelligence in Education (AIED) framework does a good job of hypothetically categorizing AI's constructivist role in the classroom. In their eyes, ChatGPT holds potential to support constructivist learning by serving as an interactive resource that allows students to strengthen their engagement through collaboration; the chatbot becomes an intelligent learning partner, able to take on the roles of tutor and tutee (Gwo-Jen Hwang & Nian-Shing Chen, 2023). The aforementioned historical relationship between constructionism and AI aligns with the idea that students should engage with ideas or materials actively - through a kind of trial and error - until a set of inputs reconstructs or refines their understanding of, not only a particular concept, but also the tool itself (Hayles 2012). By leveraging prior experiences, students work from their own pre-trained cognitive system - the neurons firing in their own brain - but also incorporate the organized unfolding of the chatbot's intelligence as well (Kohler, 2024). The tool and the student adapt to each other's decisions in an interactive environment, the effectiveness of which has to do with the GPT's power to simulate intelligent conversations and guide discussions through its dialogue management system (DMS) hardware - the “Chat” in ChatGPT.

In support of more interactive, student-led learning environments, Seymour Papert, along with fellow research Idit Harel, proposed three characteristics to undergird software design as a Learning Environment: appropriability, evocativeness, and integration (Harel & Papert,

1990) (Rospigliosi, 2023). These concepts illustrate the distinct importance of ChatGPT's interactive dialogue, where its abilities become embedded within the situated knowledge of its making: its models and parameters and algorithmic constraints (Zimmerman 1998); a deeply relational, flexible, and structural dynamic existing within a colossal lake of data. The writing process oftentimes is described as "an unending conversation" in knowledge creation. Here, ChatGPT's conversational adaptability is an authorization: students shape their own knowledge processes (appropriability) by means of tailored, personalized responses, making the information they receive their own. What's more, the conversational nature opens up students to reflection, or personal thought (evocativeness), and iterative dialogue supports the connection of new and prior knowledge (integration) (Rospigliosi, 2023). These attributes align with constructivist and student-centered learning paradigms. ChatGPT, offering opportunities for more personalized and interactive learning, allows students to deepen their understanding through the promotion of metacognitive skills (what Levine reported as "meta-level understanding") (Kim & Adlof, 2023). Through these conversational interactions, ChatGPT prompts learners to receive information, reconstruct it on a new level and surpass it, ultimately moving towards reflection and critical analysis. A few research subjects (a sophomore and a junior) from Levine's research study illustrate this clearly. The students did not copy from ChatGPT, but instead used ChatGPT's examples as a model for their own act of translation during the composing process (Levine et al., 2024). So the story goes . . .

It is possible that ChatGPT is not any of the things that some research has come to think of it as. By tracing the historical development of constructivist and constructionist learning theories alongside advancements in artificial intelligence research, my aim was to highlight their shared genealogies and examine their implications for student learning, but more so to show how the vectors of human technologies - writing, typewriters, word processors, and now, deep learning - create new possibilities in terms of knowledge-making. Specifically, I attempted to demonstrate through the research that knowledge can be constructed using this

tool, much like it is constructed through the act of writing, with both processes unfolding within a constructivist framework. There are other studies, however, that are not so pleasant. Mei Tan and Hari Subramonyam (2024) underscore ChatGPT's lack of pedagogical context. This distinction is exemplified through the example of an ELA teacher prompting ChatGPT to design a lesson plan through a constructivist lens, haplessly finding out that the pedagogical construct was present only in the language of the output rather than in the content or activities described (Tan & Subramonyam, 2024). Several other teachers prompted ChatGPT to generate writing prompts, essential questions, or comprehension questions but were disappointed in the outputs that could not distinguish the three categories of materials (Tan & Subramonyam, 2024). If a teacher struggles to prompt the chatbot, what is left to say about the attempts of our students? The algorithms or the functions within these generative pre-trained performers are only generating approximation of an object in a training set. At least for now, as Lauren Goodlad and Matthew Stone in their paper *Beyond Chatbot-K* put it: "the actual functionalities of generative AI are . . . remarkably humdrum" (Goodlad & Stone, 2024). ChatGPT's generated text, in addition to struggles with authentic engagement to the actual meaning-making activities of its real-world audience, is also ineffectual in contextualizing writing that involves challenging the status quo, or in reading or parsing against the grain to support more invigorating analysis (Goodlad & Stone, 2024). This is certainly true in the case of Tan & Subramonyam's report, as teachers noticed outputs "containing hedging and qualifying language to avoid complex stances" (Tan & Subramonyam, 2024).

Naturally this research poses serious weaknesses and limitations to the dialogic or constructivist configuration of this tool in the classroom. One of the limitations to the more dialogic theory of learning is that there isn't really a conversation to be had, there is only a statistical probability of chosen words. Students and teachers come to the chatbot looking for an explanation into how things are, but are left with questions concerning sequence prediction

(Goodlad & Stone, 2024). In other words, the linguistic communication inherent to the chatbot (the DMS hardware) is indifferent to how things really are (Trevisan et al., 2024), blatantly dismissing the importance of context in a writerly situation (Zimmerman, 1998), and as result, fails to effectively engage with a rhetorical situation to construct meaning; Consequently, it remains ignorant to the ways in which writing is shaped by place and community (Beck & Levine, 2024).

The other vision being pursued is the integration of ChatGPT through the fabric of a constructivist/constructionist lens, emphasizing the import of the tool to enhance teaching and learning over the solitary purpose of producing content. To reiterate, the goal is to augment human intelligence through specialized human-computer interaction, thereby increasing the amount of personalized and adaptive learning experiences in the classroom (Ouyang & Jiao, 2021). This approach seemingly provides a more effective means of promoting meaningful knowledge construction by simulationing processes such as solution-finding or artifact-creation (Kim & Adlof, 2023). A worthy cause. At this point, however, it remains unclear how actively students engage in knowledge creation, as they may choose to navigate the learning process unencumbered by high-friction cognitive tasks (Goodlad & Stone, 2024). Reading, planning, conducting research, translating concepts into correspondence, and eventually editing and reviewing are all time-consuming practices. The marketing of ChatGPT propagates the idea of ruthless efficiency, putting forward a deceptive ideology of “frictionless knowing” (Goodlad & Stone, 2024). The risk being that our middle school or high school students conflate the output with knowledge construction instead of enabling, amplifying, and concretizing their active role in crafting the learning process (Goodlad & Stone, 2024). Moreover, the data within OpenAI’s slush of an archive is trapped in experience – the situated knowledge of sociocultural apparatus – but this does not ensure genuine diversity. What’s at risk is the entrenchment of hegemonic norms and homogenization – a grazing over of marginalized perspectives as a result of systemic inequalities (Bender et al., 2021).

LLM-based chatbots are designed to manufacture text that persuades their customers as to their competence. If a dialogue-based tutoring system (Ouyang & Jiao, 2021) is equated with a dialogue management system (Trevisan et al., 2024), then “tutoring” becomes a misperception of collaborative dialogue. Such systems are calibrated to interact with clientele amiably, presenting a “veneer of sycophancy or professionalism” – yet they do so without fundamental regard for the truth or falsity of their claims (Trevisan et al., 2024). DON'T be alarmed, the kids and the teachers are okay. Rather than this back-and-forth inquiry versus scrutiny dichotomy, or an odyssey into the complexities surrounding self-assured professions and their counter-arguing counterparts, the main issue is how Chatgpt will trouble the traditional frameworks of the ELA classroom and the constructed knowledge-formation that takes place therein. There are critical concerns over the potential impediments to literacy, or at the very least, the challenges this tool poses to our understanding of humanistic knowledge – as something not constructed through the historical and situated nature of experience, but instead transmitted through an entity that lacks entanglement with the greater sociohistorical dimensions of thought (Higgs & Stornaiuolo, 2024). This segment began by relating how technological advancements have influenced conceptions of learning and how there exists consequential connections between artificial intelligence, constructivism, and educational technology (Hof, 2021). This segment will close as we find ourselves illustrating a foundational bifurcation: that technological advancements like ChatGPT develop as a model of human cognition or mental representations; and that this emerging model is imbricated in everyday social, cultural, economic, and political activity. What I have tried to make visible is that this technology must be understood culturally and historically, not as an isolated tool or machine but as a sociotechnical system. In other words, it should be seen or put another way, as “an entanglement of humans with each other:” the tools, the arts, financial markets, political practices, and the systematic institutionalized ways

of life, a manifestation of the contemporary condition (Higgs & Stornaiuolo, 2024) (Trevisan et al., 2024).

Encircling this tool are discussions shaped by free-market institutionalised relations – industry assertions and their close counterpart, advocacy from the state – but also perspectives from civil society, academia, and for our particular purpose, the students, the teachers, and the encompassing edifice of education writ large. This sweeping discourse oftentimes fails to capture the complex experiences of the essential players (students and teachers). With Higgs' and Stornaiuolo's study, a qualitative analysis composed to gain insights into how high school students experience and interpret ChatGPT, we see their considerate deliberations of AI-mediated writing. The heart of their concerns stemmed from genuine anxieties about underrepresentation of certain voices, bias, and plagiarism (Higgs & Stornaiuolo, 2024). This accords with some of our earlier observations, where one of Levine's subjects pointed to ChatGPT's writing coming from a linguistic space where the norms did not mesh with her sociohistorical communicative needs (Levine et al., n.d.). What the students are pointing to is that there is no public language, no semantic space in which this technical object can be plugged into (Trevisan et al., 2024). In Tan and Subramonyam's findings, teachers struggled to develop sufficient pedagogical content, encountering many limitations and inefficiencies in the wake of their maneuverings. Still, their tinkering away with their prompts in an effort to participate in ongoing critical examinations of the technology (Tan & Subramonyam, 2024). These two subjects – students and teachers – undertook ChatGPT and its modalities as both practice and a theory of that practice. It is in this way that ChatGPT is a constructivist tool. Writing with AI is more than just the act of “doing” writing, it is also the theorizing of that doing; the process of thinking about how to get something else to think (Higgs & Stornaiuolo, 2024) (Selwyn, 2017). Altogether, discussions will persist over whether the students and teachers who use ChatGPT are engaged in genuine contemplation, patterned behavior, or a mix of both. Regardless, the proposition that this new form of technology will be integral to a new formation of writing and inquiry remains. How, then will the institution of secondary education respond?

Equity, Design Thinking, and Theories of Motivation

In the discussed AIED framework (Gwo-Jen Hwang & Nian-Shing Chen, 2023), ChatGPT has surfaced as a significant influence with the potential to revolutionize learning and the educational experience more generally (Kohler, 2024). The early days of AI in Education – the ideas germinating in the 1960's and 1970's after Papert discoveries – gave rise to the field of “intelligent tutoring systems” (Sewlyn, 2017). These intelligent Tutoring Systems (ITS), briefly introduced above, lay within the space of AIED applications, as they are underlined by adaptive, personalized learning and feedback – programs that are capable of learner-friendly clarification and that are proficient in providing guidance to students through each step of a problem-solving process. According to this logic, digital technologies of this nature are able to espouse a pedagogical approach. One that appears to attend to constructivist notions of education, where the emphasis is on “activities and practices that are individually centred and individually driven” (Sewlyn, 2017). As our findings indicate, students work collaboratively with our tool, completing tasks through trial and error, interacting with the artificial neural network by integrating it into a distributed writing model; the vision being that ChatGPT's constant shedding of skins is the ultimate “one-to-one tutoring experience” (Gwo-Jen Hwang & Nian-Shing Chen, 2023) (Sewlyn, 2017). More precisely, because of its multiple roles – learning peer/partner, domain expert, learning tool, tutor or tutee – ChatGPT possesses relational significance. Students can ask it to generate ideas for *them*, organize for *them*, draft for *them*, and evaluate *their* written work (Levine et al., n.d.). AI also will, of course, change its responses within a certain encounter. For instance, a 9th grade student inputs their own prompts to create an argumentative writing outline “aligned to their individual thinking processes” (Libertino, 2024). In this case, the tool will not only provide information regarding their individual interest, but also will, depending on how it is prompted, provide additional contextual information in the greyer area running beside the prompt: the age of the student, the grade level, examples that cohere with their interests, the lesson assignment, ect. For students who might resist standardized writing outlines, or for students with disabilities who may struggle to ideate or produce writing, ChatGPT becomes another way for them to scaffold their writing process (Libertino, 2024), because the tool can, quite miraculously, mimic human memory. Students come to build and refine representations of complex systems and artifacts very much in line with the

self-directed exploratory learning central to constructivist philosophy (Sewlyn, 2017). Within this context, Papert's insights into the concurrent advancements of artificial intelligence and the imitation of childhood learning thus remain vividly relevant (Sewlyn, 2017) (Hof, 2021). Compelled by her student's frustrations and hindrances with painstakingly designed graphic organizers, Libertino writes: "I soon realized that my students needed a more personalized approach to outlining their writing." She then wonders: "why fit [the outline] into boxes when you [a student] can make them yourself?" It is interesting to note that the responsibility of the educator becomes furnishing their lesson plans with the correct prompt templates. Students, even as they may input their own prompts, engage with the AI model "via scaffolded prompting questions, tailored instructional responses, and content that is clear, concise, and uses academic vocabulary for the appropriate grade level" (Libertino, 2024). Libertino affirms that, contrary to other research, this approach is indeed possible: "Ways to tailor prompts include specifying the grade level, writing style, writing prompt, amount of paragraphs, thesis statement, and any relevant details or evidence." Some examples of which can be found [here](#). Is this a way forward, where design becomes the learning environment, and within that environment, design becomes software, and the software (and an extension of the software, the teacher) becomes the designer? In 21st century education, we have the germinating proliferation of a "design thinking" lexicon", "a nonlinear, student-centric methodology for engendering pedagogical solutions tailored to learners' needs" (Mrayhi et al., 2024). Creating inclusive and accessible learning experiences with emerging technologies is essential in today's educational landscape. The Universal Design for Learning (UDL) framework has been introduced as a strategy to address learner diversity, aligning with brain networks associated with motivation, information processing, and task execution (Mrayhi et al., 2024). In design thinking, technology becomes a medium for self-expression - a recursive loop, generated through the interaction between the technology, the teacher, and the learner, all of which are reflected through thinking with the computer. In this framework, machines come to work alongside students, empowering them to better versions of their writing selves (Harel & Papert, 1990).

In its role as a prodigious personal tutor, ChatGPT caters to a learners' individual needs and, in effect, enhances their motivation through personalized learning experiences. It is the hope that ChatGPT also helps students gauge their competence, thus boosting their motivation and development (Kohler,

2024). The self-determination theory (SDT) of motivation proposes that people have an innate drive to learn, develop mastery, and grow within their social environment. When a student's psychological needs for autonomy are met, Kohler suggests that ChatGPT can nurture a students' intrinsic motivation and thereby ignite their feelings of capability (competence), plus their sense of fulfillment (self-determination) (Kohler, 2024). More particularly, Cognitive Evaluation Theory (CET), a subset of SDT, posits that intrinsic motivations are rooted in these two drives: the psychological needs for competence and self-determination (Deci et al., 2001). Deci and his counterparts refer to the degree to which an individual perceives themselves to be in control of their own actions and decisions. Events either decrease or increase perceived self-determination or perceived competence, with a decrease in either undermining intrinsic motivation (Deci et al., 2001). Nonetheless, perceived self-determination is a prerequisite for the increase of perceived competence. These landmark findings by Deci and Co. suggestively coincide with the findings of Libertino (2024), whose research suggests that ChatGPT can influence a students' perceived locus of causality - their perception of their actions being driven by internal desires, values, and interests. As she puts it in her paper:

“[Students] no longer confined to predefined structures nor stuck staring at a blank page [...] facilitated the creation of organizers generated just for them—appropriately structured yet personalized outlines that accommodated the various ways in which they approached writing” (Libertino, 2024).

The first part of this segment deals with the concept of a software capable of delivering one-to-one tutoring. The second finding is that intrinsic motivation is a strong motivator for ChatGPT use intention, where ELA education becomes more technologically open with a focus on allowing digital literacy to foster traditional literacy. As Libertino observes, “AI became the bridge between student deficit and student success” (Libertino, 2024). The third component of our study is concerned with the student-centric methodology of Universal Design for Learning. The idea being that three main principles of UDL - engagement, representation, and action and expression - accommodate flexible learning opportunities that cater to an individuals strengths and needs. The concept of multiple means of engagement recognizes the varied motivations and engagement styles of learners, underscoring the necessity of offering diverse pathways for learners to connect with learning materials (Saborío-Taylor &

Rojas-Ramírez, 2024). In Beck and Levine's study (2024), for example, high school students were asked to write superhero stories to pitch to fictional movie companies. The session itself encouraged choice and autonomy, but students, through the use of ChatGPT, were able to optimize the relevance, value, and authenticity of their stories, with one student drawing on the name his friend group used to call themselves: The "Dorito Squad." In this case, students were able to integrate personal humor and references into ChatGPT (Beck & Levine, 2024). The bots algorithmic output spat out figurative language, "identified the word "zest" as a word that typically collocates with "Doritos," and likewise, "identified villains as common characters in superhero stories", which resulted in a villain called "Dr. Zest" (Beck & Levine, 2024). With respect to the design of multiple means of engagement, ChatGPT's abilities furthered the goal of the lesson while simultaneously sparking genuine laughter and pleasure in the writing process (CAST, 2024), or what Beck and Levine (2024) call "authentic delight."

Subsequently, multiple means of representation focuses on aspects of learning by acknowledging that learners differ in how they perceive and understand information (Saborío-Taylor & Rojas-Ramírez, 2024). This raises questions about the platform's abilities to present information in more ways than one. Of course, it can clarify complex terms or expressions and decode or compose a simpler word summary of a text, which allows students to approach content through their individual learning or proficiency level (CAST, 2024). This view is supported by Beck and Levine (2024), who write that "one pair of students was stuck on finding a synonym for the word 'suitable.' They asked ChatGPT for alternatives [...] after some discussion, the students determined that 'felicitous' was the word they wanted and drafted a sentence" (Beck & Levine, 2024). In line with a multiple means of representation format, ChatGPT's flexibility establishes linguistic comprehension and skill-building, with the students determining the degree of assistance they need while, hopefully, evaluating that assistance with a critical eye (Beck & Levine, 2024). Lastly, embracing multiple means of action and expression means offering a spectrum of assessment methods: written assignments, oral presentations, or project-based assessments to suit diverse communication, preferences, and abilities (Saborío-Taylor & Rojas-Ramírez, 2024). With the right inputs and linguistic cues, ChatGPT's algorithm searches its database for a way to express knowledge in differentiated styles. As Higgs and Stornaiuolo (2024) put it, ChatGPT invites students to "catalyze their thinking and writing processes," supplementing their composition with AI as a "thought partner" (Higgs & Stornaiuolo, 2024). While multiple means of

action or expression may focus more closely on the actual expression or attainment of a learning goal, ChatGPT acts as an instrument to interaction and action. This can be illustrated briefly by the students in Higgs and Stornaiuolo's study, who describe AI more broadly as "a catalyst for my thoughts", "a spark to go off from," and "ideas to go off of for my writing" (Higgs & Stornaiuolo, 2024). Put simply, learning becomes more efficiently accessible through interaction with the tool, leaving students to leverage their unique skills and perspectives in composition (CAST, 2024). The collaboration between UDL and AI are believed to promote this ideas of self-directed learning, or autonomous learning; the hope being that by embracing digital technologies in education, educators or administrators of learning begin the rather ambitious move towards removing barriers to development, ensuring accessibility for all students (Saborío-Taylor & Rojas-Ramírez, 2024). Teachers take on a key role in guiding students towards intentional prompt planning, and thus the deepening of their metalevel knowledge of writing. This will require teachers to build familiarity with AI prompting as a means to support students in the design and evaluation of their results (Beck & Levine, 2024). This partnership ultimately adopts a social justice lens that configures some type of revolutionary potential between artificial intelligence and educational systems. By promoting inclusivity and autonomous learning through design thinking strategies - addressing a students evolving needs and crafting adaptive educational experiences - it aims to challenge and disrupt systemic inequities (Saborío-Taylor & Rojas-Ramírez, 2024) (Mrayhi et al., 2024).

A cherished development for computer scientists and educational technology developers is a software with the capabilities of delivering one-to-one tutoring. Above, ChatGPT is proposed as an intelligent tutor that can provision a young writer's individual needs and enhance their motivation through personalized learning experiences. The verdict, through a framework like AIED, is officially out: ChatGPT is here to serve - a digital assistance par excellence, akin to Siri. It is vital to note that the tension between opposing perspectives - those of technophiles advocating for the unfettered adoption of AI and those of technophobes warning against its potential risks - reflects a critical gap in public understanding (Pasquinelli, 2019). This portion of the paper is concerned with examining the uses and effects of ChatGPT through the principles of motivation and UDL paradigm previously mentioned, and assessing the ever growing costs and risks associated with the pursuit of it. To start, its mimicry may

afford human-centered forms of deviation that show potential to aid in learner autonomy, yet there are still plenty of surfacing problems, as previous research has tried to make clear, including gaps and omissions in training data, amplification of bias, and misinformation (Goodlad & Stone, 2024). The vast, uncuration training datasets used to build these models masks biases and limitations under the guise of objectivity and precision (Pasquinelli, 2019). One outcome from a ChatGPT interaction experiment illustrates the issue; James H. Oldham, in a paper aptly titled, *ChatGPT: The Co-Teacher We Need?* considers the potential for *user interaction bias*, in which his writing students selectively steer their work by posing leading questions that reinforce their previously conceived notions, instead of conducting thorough research of the literature to develop informed conclusions. This can prevent students from seeking out new information that may challenge their beliefs and endangers a validation of a limited set of ideas (Oldham, 2024). While prior work with ChatGPT has primarily focused on preventing models from generating harmful content, there is an increasing need for language models to avoid complying with every user request (Brahman et al., 2024). A closer elucidation indicates that “for underspecified requests and requests with false presuppositions, the models tend to assume user-intent and directly answer questions without asking for any clarification” (Brahman et al., 2024). These issues may fortify the leading questions that prevented Oldham’s students from informed conclusions that incorporated a diversity of perspectives. What’s more, Pasquinelli (2019) notes that these machine learning technologies impose a process of normalization by way of information compression, in what is known as dimensionality reduction. In the pursuit of reducing calculation costs, machine learning systems often aggregate and eliminate low-variance categories – those whose values barely fluctuate. This process of dimensionality reduction streamlines data, and collapses cultural taxonomies along the way. Emerging through the ruins is a troubling form of category reduction: the compression of diversity into a bland, homogenized norm. Eventually, Pasquinelli (2019) argues that the effect of machine learning on world diversity is normalisation; “that is, an equalisation of anomalies to an average norm.” Having said that, in multiple studies students recognized an imperative to resist the flat, mechanical rewriting offered by the bot, favoring their own language and voice instead. Oldham’s (2024) examination, which instructed students to brainstorm their own narrative prompts and then evaluate ChatGPT’s two-page narratives based on them. Students

constantly noted that the ChatGPT-generated narratives were unclear and lacked a sense of "realness," as they were often mechanical and formulaic. A common response from students was the evaluation of writing as a way of creating personal knowledge; students saw writing as a way to "connect with their community", and further perceived ChatGPT as an outsider to their communities (Beck & Levine, 2024). In a related study, a response from a young student - who identified as Black and chose not to use AI - alluded to the potential connections between AI usage and broader racial, social, or socioeconomic factors, stating, "What I see is that people like me doesn't use it a lot" (Higgs & Stornaiuolo, 2024). Whether as ancient as written language or as contemporary as AI-mediated writing, working with a technology requires contextualizing it within the social and cultural domains of its creation and use (Higgs & Stornaiuolo, 2024) (Zimmerman, 1998). Going forward, students may tend to adapt their behaviors to algorithms that intensify standardized perspectives, thereby worsening systemic inequalities (Bender et al., 2021) (Pasquinelli, 2019). The reimagining of our educational system calls for the visionary design of inclusive, student-centered learning. Systems like ChatGPT are marketed as delivering the ultimate "one-to-one" tutoring experience, facilitating greater student autonomy as they discover and construct knowledge independently. With the adoption of an "optimize teaching and learning for all" UDL strategy and the growing recognition of a young person's basic psychological and motivational needs, students are increasingly positioned as users. In such an environment, where does control ultimately reside?

That a student must perceive, and only perceive, their actions stemming from internal desires, values, and interests is a fascinating viewpoint in which to continue a review of our tool. We must remember that intrinsic motivation depends on perceptions of competence and self-determination. Events that restrict autonomy or impose control tend to weaken intrinsic motivation, while those that promote choice, ownership, and control can increase self-determination. Similarly, events that increase perceived competence will enhance intrinsic motivation, as long as they are accompanied by perceived self-determination. ChatGPT is said to enhance perceived self-determination, as students often find satisfaction in its perceived usefulness for accomplishing extrinsic tasks as it gives them more self-sufficiency

and, importantly, efficiency, when stimulating their thinking and writing processes (Higgs & Stornaiuolo, 2024). By providing informational feedback, or information that reinforces a students self-determined competence, Kohler (2024) suspects GenAI like ChatGPT can promote motivation and development. Frameworks like AIED conceptualize these possibilities as transformative, envisioning significant educational change (Gwo-Jen Hwang & Nian-Shing Chen, 2023). Coinciding with the rise of these technosolutionist ideas, this era has also introduced the possibility that everything could in fact be different. While ChatGPT is capable of facilitating self-directed learning - or at least self-directed pursuit - it primarily serves as a tool to accelerate the achievement of extrinsic rewards; Hence, it constitutes an augmentation of the two-step process that drives intrinsic motivation. One of the more important drawbacks of the tool in the writing process lies in the growing reliance on it. For Students, remarks Oldham (2024): "this is as easy as copying assignment prompts into ChatGPT's text box and receiving a usually coherent and acceptable response to submit as their own." In this respect, students become passive users rather than active creators, a thesis that dismisses the value of intrinsic motivation for these students but retains the perception of it. A core issue lies in the marketing hook, which hinges on the premise that writing is inherently a waste of time; the chatbotization of writing as "not process but product, not provenance but output" (Goodlad & Stone, 2024). The writing process involves planning, which involves generating an internal representation of knowledge and organizing ideas. It also involves translating, which transforms this internal representation into written language (Zimmerman, 1998). In between these two states there is friction: "the experience of difficulty, resistance, and the passage of time [...] an inescapable index of the condition of being conscious, embodied, interactive, and alive (Goodlad & Stone, 2024). We could imagine a world where our chatbot effectively acts serves as a catalyst for students to exercise control over their learning and academic potential. But is that this world? This perspective requires us to reflect on how *perception* shapes a students understanding of intrinsic motivation. One implication of this analysis is that certain technological affordances may contribute to the "privation" or absence of human flourishing. By impoverishing human capacities, technological affordances have the power to erode skill, weaken strength, and fragment attention, eventual diminishing the capacities that make us human (Ferdman, 2023). In contrast to high-friction cognitive tasks - where writing and reading

serve to transform connections into knowledge-creation – the process of “datafiction” operates on the assertion that inputting a source file and scraping its contents is equivalent to the act of reading or writing an essay. (Goodlad & Stone, 2024). Interacting with ChatGPT and each other, some student participants in Beck and Levine’s study unquestionably attributed human cognition to the bot, referring to its outputs as “facts” (Beck & Levine, 2024). ChatGPT invites its users to regard its outputs as valid, particularly when a topic of a text is unfamiliar (Beck & Levine, 2024) (Cheng et al., 2022). This reliance shifts causality to an external source, often unconsciously, subverting intrinsic motivation overtime. The result is a diminished ability to engage effectively in critical cognitive tasks, such as perceiving, remembering, and planning (Ferdman, 2023). The competent exercise of one’s capacities, coupled with the production of an intrinsically valuable output, is what we define as human flourishing (Ferdman, 2023). As Ferdman (2023) aptly observes “being dropped off at the summit by a helicopter does not constitute the same valuable outcome as actually making the climb” (Ferdman, 2023). In *Beyond CheatBots* (2024), Christopher Mah identifies that in both student and teacher discussions of ChatGPT, the central question is to whether the cognitive effort was primarily undertaken by the student or delegated to the chatbot. The unresolved tensions as to whether ChatGPT is a shortcut or scaffold to the writing process becomes more of a question as to whether we would like to move from typing to clicking, and what this means for the underdevelopment and underutilization of our human capacities. Avigaill Ferdman (2023) explains that a text that is generated by ChatGPT holds no intrinsic value to the user because it does not involve the exercising of creativity, describing this dynamic as a tool for ‘harmful scaffolding.’ If ChatGPT reduces the motivational urge to pursue knowledge as an epistemic good, it undermines the user’s interest in development. Ferdman’s contention is that students navigate a social environment where falling behind academically means falling into economic precarity. In an educational setting, the perceived appearance of self-determination and competence that underpin intrinsic motivation preoccupies students with signs of success over substantive value. As a result, the long-standing purpose of education – to shape individuals through structured learning – begins to unravel before our eyes. The key to understanding what is concealed is not only the affordances provided by the tool itself, but the “political economy it operates in, and the norms that will be ingrained around it.”

Transformative Potential?

Conclusion

Bibliography

Beck, S. W., & Levine, S. (2024). The Next Word: A Framework for Imagining the Benefits and Harms of Generative AI as a Resource for Learning to Write. *Reading Research Quarterly*, 59(4), 706–715. <https://doi.org/10.1002/rrq.567>

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? 🦜. *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. <https://doi.org/10.1145/3442188.3445922>

Blackwell, A. F. (2024). *Moral codes: Designing alternatives to AI*. The MIT Press. <https://doi.org/10.7551/mitpress/14872.001.0001>

Brahman, F., Kumar, S., Balachandran, V., Dasigi, P., Pyatkin, V., Ravichander, A., Wiegrefe, S., Dziri, N., Chandu, K., Hessel, J., Tsvetkov, Y., Smith, N. A., Choi, Y., & Hajishirzi, H. (2024). The Art of Saying No: Contextual Noncompliance in Language Models (No. arXiv:2407.12043). arXiv. <https://doi.org/10.48550/arXiv.2407.12043>

Brown, T. B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D. M., Wu, J., Winter, C., ... Amodei, D. (2020). Language models are few-shot learners (No. arXiv:2005.14165). arXiv. <https://doi.org/10.48550/arXiv.2005.14165>

CAST. (2024). *About universal design for learning*. <https://www.cast.org/impact/universal-design-for-learning-udl>

Cheng, R., Smith-Renner, A., Zhang, K., Tetreault, J. R., & Jaimes, A. (2022). Mapping the Design Space of Human-AI Interaction in Text Summarization (No. arXiv:2206.14863). arXiv. <https://doi.org/10.48550/arXiv.2206.14863>

Deci, E. L., Koestner, R., & Ryan, R. M. (2001). Extrinsic Rewards and Intrinsic Motivation in Education: Reconsidered Once Again. *Review of Educational Research*, 71(1), 1–27. <https://doi.org/10.3102/00346543071001001>

Eisenstein, J. (2019). *Introduction to natural language processing*. The MIT Press.

Erickson, K. (2023, April 10). What a precursor to ChatGPT taught us about AI — in 1966. *Salon*. <https://www.salon.com/2023/04/10/what-a-precursor-to-chatgpt-taught-us-about-ai--in-1966/>

Ferdman, A. (2023). Human Flourishing and Technology Affordances. *Philosophy & Technology*, 37(1), 1. <https://doi.org/10.1007/s13347-023-00686-9>

Goodlad, L. M. E., & Stone, M. (2024). Beyond Chatbot-K: On Large Language Models, “Generative AI,” and Rise of Chatbots—An Introduction. *Critical AI*, 2(1).
<https://doi.org/10.1215/2834703X-11205147>

Gwo-Jen Hwang, & Nian-Shing Chen. (2023). Editorial position paper: Exploring the potential of generative artificial intelligence in education: Applications, challenges, and future research directions. *Educational Technology & Society*, 26(2), 1–18.
[https://doi.org/10.30191/ETS.202304_26\(2\).0014](https://doi.org/10.30191/ETS.202304_26(2).0014)

Harel, I., & Papert, S. (1990). Software Design as a Learning Environment. *Interactive Learning Environments*, 1(1), 1–32. <https://doi.org/10.1080/1049482900010102>

Hayles, N. K. (1999). *How we became posthuman*. University of Chicago Press.

Hayles, N. K. (2012). *How we think: Digital media and contemporary technogenesis*. The University of Chicago Press.

Introducing ChatGPT. (n.d.). Retrieved November 18, 2024, from
<https://openai.com/index/chatgpt/>

Higgs, J. M., & Stornaiuolo, A. (2024). Being Human in the Age of Generative AI: Young People’s Ethical Concerns about Writing and Living with Machines. *Reading Research Quarterly*, 59(4), 632–650. <https://doi.org/10.1002/rrq.552>

Hof, B. (2021). The turtle and the mouse: How constructivist learning theory shaped artificial intelligence and educational technology in the 1960s. *History of Education*, 50(1), 93–111.

<https://doi.org/10.1080/0046760X.2020.1826053>

Kahn, K., & Winters, N. (2021). Constructionism and AI: A history and possible futures. *British Journal of Educational Technology*, 52(3), 1130–1142. <https://doi.org/10.1111/bjet.13088>

Kak, A., & Myers West, S. (2023). AI Now 2023 landscape: Confronting tech power. *AI Now Institute*. <https://ainowinstitute.org/2023-landscape>

Kelleher, J. D. (2019). *Deep learning*. The MIT Press.

<https://doi.org/10.7551/mitpress/11171.001.0001>

Kim, M., & Adlof, L. (2023). Adapting to the Future: ChatGPT as a Means for Supporting Constructivist Learning Environments. *TechTrends*, 68, 1–10.

<https://doi.org/10.1007/s11528-023-00899-x>

Kohler, K. (2024). You Only Need to Change Your Direction: A Look at the Potential Impact of ChatGPT on Education. *Technology in Language Teaching & Learning*, 6(1).

<https://eric.ed.gov/?id=EJ1444002>

Levine, S., Beck, S. W., Mah, C., Phalen, L., & Plttman, J. (n.d.). How do students use ChatGPT as a writing support? *Journal of Adolescent & Adult Literacy*, n/a(n/a).

<https://doi.org/10.1002/jaal.1373>

Libertino, M. (2024). Reimagining ELA: Artificial Intelligence as a Literacy Intervention Tool. *English Leadership Quarterly (Online)*, 47(1), 19–25.

Mah, C., Walker, H., Phalen, L., Levine, S., Beck, S. W., & Pittman, J. (2024). Beyond CheatBots: Examining Tensions in Teachers' and Students' Perceptions of Cheating and Learning with ChatGPT. *Education Sciences*, 14(5), Article 5.

<https://doi.org/10.3390/educsci14050500>

Mrayhi, S., Khribi, M. K., Belhadj, H., & Jemni, M. (2024). Designing Future Education for All: Principles and Frameworks. In R. Huang, D. Liu, M. A. Adarkwah, H. Wang, & B. Shehata (Eds.), *Envisioning the Future of Education Through Design* (pp. 147–177). Springer Nature.

https://doi.org/10.1007/978-981-97-0076-9_7

Oldham, J. H. (2024). ChatGPT: The Co-teacher We Need? *English Journal*, 113(4), 53–61.

Ouyang, F., & Jiao, P. (2021). Artificial intelligence in education: The three paradigms. *Computers and Education: Artificial Intelligence*, 2, 100020.

<https://doi.org/10.1016/j.caeai.2021.100020>

Pasquinelli, M. (2019). HOW A MACHINE LEARNS AND FAILS – A GRAMMAR OF ERROR FOR ARTIFICIAL INTELLIGENCE.

Rospigliosi, P. 'Asher.' (2023). Artificial intelligence in teaching and learning: What questions should we ask of ChatGPT? *Interactive Learning Environments*, 31(1), 1–3.

<https://doi.org/10.1080/10494820.2023.2180191>

Rudnicka, K. (2023). Can Grammarly and ChatGPT accelerate language change? AI-powered technologies and their impact on the English language: Wordiness vs. conciseness.

Procesamiento de Lenguaje Natural, 71, 205–214.

Saborío-Taylor, S., & Rojas-Ramírez, F. (2024). Universal design for learning and artificial intelligence in the digital era: Fostering inclusion and autonomous learning. *International Journal*

of Professional Development, Learners and Learning, 6(2), ep2408.

<https://doi.org/10.30935/ijpdll/14694>

Selwyn, N. (2017). *Education and technology: Key issues and debates* (2nd ed.). Bloomsbury Academic.

Shien, V. T. C. (2024). A sociological take on language generative AI tools. *International Journal of Emerging Technologies in Learning (IJET)*, 19(07), Article 07.

<https://doi.org/10.3991/ijet.v19i07.50853>

Tan, M., & Subramonyam, H. (2024). More than Model Documentation: Uncovering Teachers' Bespoke Information Needs for Informed Classroom Integration of ChatGPT. Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, 1–19.

<https://doi.org/10.1145/3613904.3642592>

Tarnoff, B. (2023, July 25). Weizenbaum's nightmares: How the inventor of the first chatbot turned against AI. *The Guardian*.

<https://www.theguardian.com/technology/2023/jul/25/joseph-weizenbaum-inventor-eliza-chatbot-turned-against-artificial-intelligence-ai>

Trevisan, A., Giddens, H., Dillon, S., & Blackwell, A. F. (2024). Measuring Bullshit in the Language Games played by ChatGPT (No. arXiv:2411.15129). arXiv.

<https://doi.org/10.48550/arXiv.2411.15129>

Vipra, J., & Korinek, A. (2023). Market concentration implications of foundation models (No. arXiv:2311.01550). arXiv. <https://doi.org/10.48550/arXiv.2311.01550>

Weatherby, L., & Justie, B. (2022). Indexical AI. *Critical Inquiry*, 48(2), 381–415.

<https://doi.org/10.1086/717312>

Zimmerman, B. B. (1998). Linda Flower and social cognition: Constructing a view of the writing process. *ACM SIGDOC Asterisk Journal of Computer Documentation*, 22(3), 25–37.

<https://doi.org/10.1145/381808.381817>